THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 13: **Our Environment** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In which of the following food chains is the energy available to the final consumer the greatest? 1

(a) Plankton, fish, bird **(b)** Grass, grasshopper, frog, snake, eagle
(c) Grass, insect, bird, snake **(d)** Grass, goat, human

2. CFC-free refrigerators were developed to 1

(a) save electricity only **(b)** protect the ozone layer
(c) increase noise **(d)** add more cooling

3. Assertion (A): The ozone layer protects life on the earth. Reason (R): It absorbs the harmful ultraviolet radiation of the Sun. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. If the population of herbivores in a grassland dropped sharply, the immediate effects are 1

(a) grass decreases and carnivores increase **(b)** grass increases and carnivores decrease
(c) both decrease **(d)** both increase

5. Which of the following is the best way to dispose of kitchen waste? 1

(a) Burning **(b)** Dumping in drains
(c) Throwing in a river **(d)** Composting

6. Assertion (A): CFCs are widely used as refrigerants. Reason (R): CFCs are the main reason for the depletion of ozone. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A top carnivore receives 1 J of energy. The energy of the producers in a four-level food chain would be about 1

(a) 10,000 J **(b)** 1000 J **(c)** 100 J **(d)** 10 J

8. An environmentally sound action is 1

(a) using paper bags and reusing bottles **(b)** using aerosols
(c) dumping waste in drains **(d)** burning polythene

9. **Assertion (A):** The number of trophic levels in a food chain is limited to three or four. 1
Reason (R): About 90 percent of the energy is passed on from one trophic level to the next.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The ozone layer is formed by the action of infrared radiation on oxygen. 1
Reason (R): Ozone absorbs the harmful ultraviolet radiation of the Sun.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is meant by decomposers acting as the 'nature's recyclers'? 2

12. Why is the biodegradable waste disposed of better by composting than by burning? 2

13. Why is it unsafe to feed on the top level of the food chain of a polluted lake? 2

14. One per cent of the sunlight falling on a field, 1,000,000 J, is captured by plants. Find the energy at the herbivore level (use the 10 per cent law). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A forest has been cleared and replaced by a crop field. Compare the two ecosystems in terms of the number of species, food web and stability. 3

16. If the pesticide DDT is banned today, why will it continue to be found in organisms for some years? 3

17. What would happen to the grassland food chain if the carnivores were all killed? 3

OR

Why are decomposers important even though they are not part of the grazing food chain? 3

18. Why is the 10 per cent of energy only an average value? 3

OR

A student claims 'Energy is recycled in an ecosystem'. Give two arguments to correct this. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a farming village in Punjab, farmers sprayed the pesticide DDT on crops. Later, tests showed a small amount of DDT in the grains and in the bodies of insects that fed on them, a higher amount in the frogs that ate those insects and the highest amount in the snakes and the birds of prey that ate the frogs. 5
(i) Name the phenomenon of increasing concentration of harmful chemicals at higher trophic levels.
(ii) Which organisms have the highest concentration?
(iii) Why do these chemicals accumulate?
(iv) Suggest two measures to reduce this problem.

20. Describe the sequence of events by which CFCs cause ozone depletion, the consequences, and the international measures taken. Also state two actions that individuals can take. 5

OR

Discuss how the following actions affect the environment and the organisms: 5
(a) burning plastic
(b) using CFC-based sprays
(c) dumping sewage in rivers
(d) overuse of pesticides. Give one remedy for each.

END OF QUESTION PAPER — Total: 40 Marks